ETALE COVERS INDUCE ETALE REPRESENTABLE MORPHISMS OF RELATIVE HILBERT STACKS

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span; proof complete as written.

Allowed standard prerequisites (editorial).

- Section functor. lemma-space-of-sections (same chapter, named). Provides the etale algebraic space F of sections of W -> Z.
- Surjectivity of sections. lemma-surjection-space-of-sections (same chapter, named). Surjectivity when G is surjective.

Source reference. Original lemma 05XN: sources/stacks-criteria/pinned-original-source.tex, source lines 1944–1944. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 05XR: sources/stacks-criteria/pinned-original-source.tex, source lines 1379–1379. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 05XQ: sources/stacks-criteria/pinned-original-source.tex, source lines 1321–1321. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *In the situation of Lemma 05XN assume*

(1) $\mathcal{Y}' = \mathcal{Y}$ *and* $H = id_{\mathcal{Y}}$,
(2) G *is representable by algebraic spaces and étale.*

Then $\mathcal{H}_d(\mathcal{X}'/\mathcal{Y}) \to \mathcal{H}_d(\mathcal{X}/\mathcal{Y})$ *is representable by algebraic spaces and étale. If* G *is also surjective, then* $\mathcal{H}_d(\mathcal{X}'/\mathcal{Y}) \to \mathcal{H}_d(\mathcal{X}/\mathcal{Y})$ *is surjective.*

Proof. Let U be a scheme and let $\xi = (U, Z, y, x, \alpha)$ be an object of $\mathcal{H}_d(\mathcal{X}/\mathcal{Y})$ over U. We have to prove that the 2-fibre product

$$(Sch/U)_{fppf} \times_{\xi, \mathcal{H}_d(\mathcal{X}/\mathcal{Y})} \mathcal{H}_d(\mathcal{X}'/\mathcal{Y}) \tag{0.1.1}$$

is representable by an algebraic space étale over U. An object of this over U' corresponds to an object x' in the fibre category of $\mathcal{X}'$ over $Z_{U'}$ such that $G(x') \cong x|_{Z_{U'}}$. By assumption the 2-fibre product

$$(Sch/Z)_{fppf} \times_{x, \mathcal{X}} \mathcal{X}'$$

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is representable by an algebraic space W such that the projection $W \to Z$ is étale. Then (0.1.1) is representable by the algebraic space F parametrizing sections of $W \to Z$ over U introduced in Lemma 05XR. Since $F \to U$ is étale we conclude that $\mathcal{H}_d(\mathcal{X}'/\mathcal{Y}) \to \mathcal{H}_d(\mathcal{X}/\mathcal{Y})$ is representable by algebraic spaces and étale. Finally, if $\mathcal{X}' \to \mathcal{X}$ is surjective also, then $W \to Z$ is surjective, and hence $F \to U$ is surjective by Lemma 05XQ. Thus in this case $\mathcal{H}_d(\mathcal{X}'/\mathcal{Y}) \to \mathcal{H}_d(\mathcal{X}/\mathcal{Y})$ is also surjective. $\square$